

12.1 Three-Dimensional Coordinate Systems

1 Warm Up

Problem 1: Find the distance between (x_1, y_1) and (x_2, y_2) in two dimensions. Find the midpoint.

Problem 2: What *is* a circle? What is the equation of a circle centered at (h, k) and radius r ?

2 Motivation

We live in a 3-D world. When we say "3"-dimensions we mean that we can represent any point in our 3-D space with three numbers (a, b, c) . There are actually infinite ways we could set up a 3-D coordinate system (just like in 2-D) but if we were to create it ourselves, what would be the best way to do this? Can we see that a lot of what happens in 3-D is analogous to 2-D?

3 Definitions & Theorems

1. Rectangular coordinate system.
2. Right hand rule.
3. Octants (Why are there 8?).
4. Projections. $(a, b, 0), (a, 0, c), (0, b, c)$.
5. The **distance** between (x_1, y_1, z_1) and (x_2, y_2, z_2) is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$
6. The **midpoint** of (x_1, y_1, z_1) and (x_2, y_2, z_2) is $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2})$.
7. The equations of a **sphere** with center (h, k, l) and radius r is $(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$.

4 Examples

Problem 1. *What does the surface $z = -2$ and $y = 3$ look like? Which points satisfy $y = x^2 + 1$ and $z = 4$?*

Problem 2. *Find the distance between $(2, -6, -9)$ and $(-1, 1, 3)$. Find the midpoint.*

Problem 3. *Find the equation of the sphere with two endpoints of a diameter at $(1, -3, -5)$ and $(-2, -2, -6)$.*

Problem 4. *Show that $x^2 + y^2 + z^2 - 6x + 12y - z + 3 = 0$ is the equation of a sphere. Find its center and radius.*

5 Scratch Work